



AJKD Atlas of Renal Pathology: Fabry Nephropathy

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Clinical and Pathologic Features

Clinical manifestations are heterogeneous. Classic cases present with peripheral neuropathy, skin vascular dilatations (angiokeratoma corporis diffusum), corneal opacities, abdominal pain and diarrhea, arrhythmias, ventricular hypertrophy, strokes, and kidney failure. The initial clinical manifestations of kidney involvement are microalbuminuria and proteinuria, which may start in the teenage years. This is followed by progressive glomerular filtration rate decline. Untreated patients (especially men) develop end-stage renal disease in their 40s and 50s.

Light microscopy: The characteristic accumulated glycosphingolipid (GSL) inclusions are removed during tissue processing for paraffin embedding. Therefore, the cells, especially podocytes, parietal epithelial cells, and distal tubular epithelial cells, appear vacuolated. Hyaline-like material accumulates in the media of arteries and arterioles (Fabry arteriopathy) and sometimes in the mesangial regions.

Immunofluorescence: Negative.

Electron microscopy: Highly electron-dense multilamellar inclusions of GSL (myelin figures made of concentric layers and zebra bodies, which have elongated striped appearance) are present in various cell types. The inclusions also stain darkly by toluidine blue on semi-thin sections. Abundant inclusions are present in podocytes, parietal epithelial cells, and distal tubular cells. Mesangial and endothelial cell inclusions may not be present in patients treated with enzyme replacement therapy or in women who are disease carriers. Arteries and arterioles show inclusions in smooth muscle cells and hyaline-like material, consistent with Fabry arteriopathy.

Etiology/Pathogenesis

Fabry disease is an X-linked lysosomal storage disease due to deficiency of alpha-galactosidase A.

This results in accumulation of GSLs, especially globotriaosylceramide (GL3), in the form of intralysosomal inclusions.

Differential Diagnosis

Cytoplasmic inclusions similar to Fabry inclusions can also be seen in iatrogenic phospholipidosis secondary to amphophilic drugs, such as chloroquine, amiodarone, and hydroxychloroquine. Foamy podocytes by light microscopy can also be seen in other lysosomal storage diseases. However, inclusions in those conditions appear different from Fabry inclusions by electron microscopy.

Key Diagnostic Features

- Foamy podocytes and distal tubular cells on paraffin sections by light microscopy
- Darkly stained round multilamellar inclusions on toluidine blue-stained sections
- Intracellular inclusions (myelin figures and zebra bodies), with greatest abundance in podocytes by electron microscopy

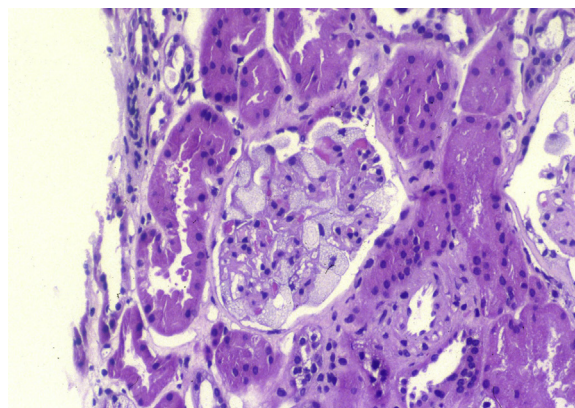


Figure 1. Fabry nephropathy with prominent vacuolization of podocytes (hematoxylin and eosin stain). Reproduced with permission from *AJKD* 37(4):e25.

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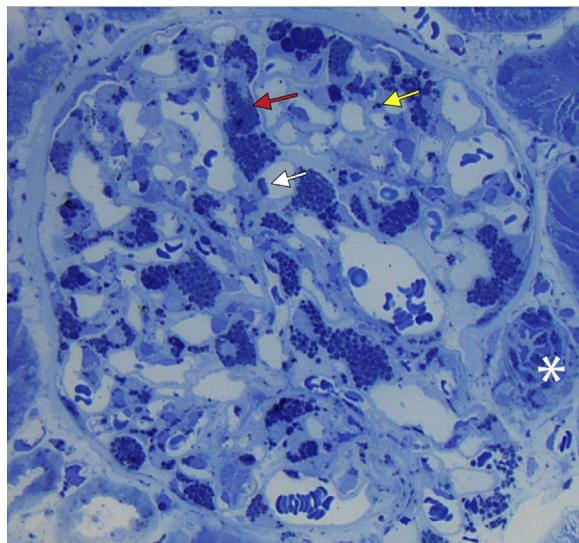


Figure 2. Fabry nephropathy with darkly stained round inclusions in podocytes (red arrow), mesangial cells (yellow arrow), and endothelial cells (white arrow) in a glomerulus. There is an arteriole with Fabry inclusions in smooth muscle cells (asterisk; toluidine blue stain).

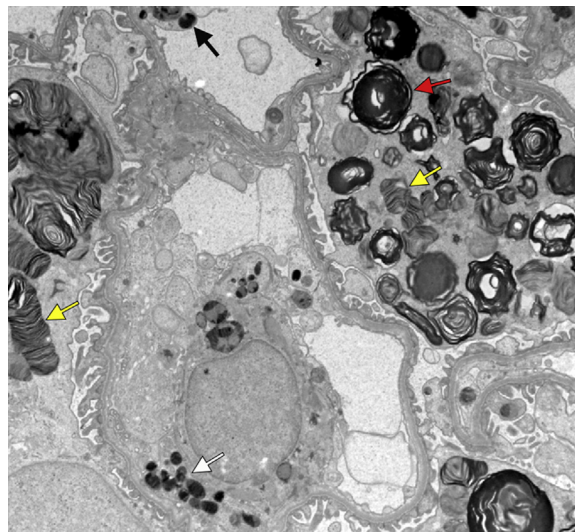


Figure 4. Fabry nephropathy with typical Fabry glycosphingolipid inclusions shaped as multilamellated myelin figures (red arrow) and zebra bodies (yellow arrows) in podocytes. There are also smaller inclusions in endothelial cells (black arrow) and mesangial cells (white arrow; electron microscopy).

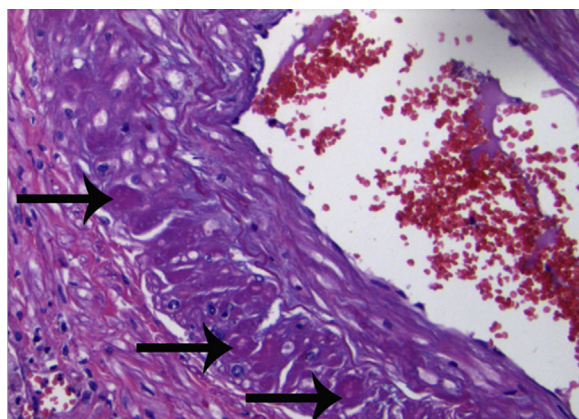


Figure 3. Fabry nephropathy with deposition of hyaline-like material in the media of an interlobular artery (arrows), consistent with Fabry arteriopathy (periodic acid–Schiff stain).